MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear functions

05

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 D

D. 9,600,000

Choice D is correct. Let y represent the number of cells per milliliter x hours after the initial observation. Since the number of cells per milliliter doubles every 3 hours, the relationship between x and y can be represented by an exponential equation of the form $y = ab^{\frac{x}{k}}$, where a is the number of cells per milliliter during the initial observation and the number of cells per milliliter increases by a factor of b every k hours. It's given that there were 300,000 cells per milliliter during the initial observation. Therefore, $a = 300{,}000$. It's also given that the number of cells per milliliter doubles, or increases by a factor of 2, every 3 hours. Therefore, $b = 2$ and $k = 3$. Substituting 300,000 for a, 2 for b, and 3 for k in the equation $y = ab^{\frac{x}{k}}$ yields $y = 300{,}0002^{\frac{x}{3}}$. The number of cells per milliliter there will be 15 hours after the initial observation is the value of y in this equation when $x = 15$. Substituting 15 for x in the equation $y = 300{,}0002^{\frac{x}{3}}$ yields $y = 300{,}0002^{\frac{15}{3}}$, or $y = 300{,}0002^5$. This is equivalent to $y = 300{,}00032$, or $y = 9{,}600{,}000$. Therefore, 15 hours after the initial observation, there will be 9,600,000 cells per milliliter.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2

14

Accept also: −5 , −4

The correct answer is either 14, -5, or -4. The x-intercepts of a graph in the xy-plane are the points at which the graph intersects the x-axis, or when the value of y is 0. Substituting 0 for y in the given equation yields $0 = 3x - 14x + 5x + 4$. Dividing both sides of this equation by 3 yields $0 = x - 14x + 5x + 4$. Applying the zero product property to this equation yields three equations: $x - 14 = 0$, $x + 5 = 0$, and $x + 4 = 0$. Adding 14 to both sides of the equation $x - 14 = 0$ yields $x = 14$, subtracting 5 from both sides of the equation $x + 5 = 0$ yields $x = -5$, and subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the x-coordinates of the x-intercepts of the graph of the given equation are 14, -5, and -4. Note that 14, -5, and -4 are examples of ways to enter a correct answer.

Q3

410

The correct answer is 410. It's given that t minutes after an initial observation, the number of bacteria in a population is $60{,}0002^{\frac{t}{410}}$. This expression consists of the initial number of bacteria, 60,000, multiplied by the expression $2^{\frac{t}{410}}$. The time it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{410}}$ to double. Since the base of the expression $2^{\frac{t}{410}}$ is 2, the expression $2^{\frac{t}{410}}$ will double when the exponent increases by 1. Since the exponent of the expression $2^{\frac{t}{410}}$ is $\frac{t}{410}$, the exponent will increase by 1 when t increases by 410. Therefore the time, in minutes, it takes for the number of bacteria in the population to double is 410.

Q4

D

D. The estimated data traffic, in terabytes, in 2010

Choice D is correct. Since t represents the number of years after 2010, the estimated data traffic, in terabytes, in 2010 can be calculated using the given equation when $t = 0$. Substituting 0 for t in the given equation yields $D = 5{,}640(1.9)^0$, or $5{,}640(1) = 5{,}640$. Thus, 5,640 represents the estimated data traffic, in terabytes, in 2010.

Choice A is incorrect. Since the equation is exponential, the amount of increase of data traffic each year isn't constant. Choice B is incorrect. According to the equation, the percent increase in data traffic each year is 90%. Choice C is incorrect. The estimated data traffic, in terabytes, for the year that is t years after 2010 is represented by D, not the number 5,640.

Q5

B

B. 5

Choice B is correct. It's given that the graph models the number of active projects a company was working on x months after the end of November 2012. Therefore, the value of x that corresponds to the end of November 2012 is 0. The point at which $x = 0$ is the y-intercept of the graph. It follows that the y-intercept of the graph shown is the point $0, 5$. Therefore, according to the model, the predicted number of active projects the company was working on at the end of November 2012 is 5.

Choice A is incorrect. This is the value of x that corresponds to the end of November 2012, not the predicted number of active projects the company was working on at the end of November 2012.

Choice C is incorrect. This is the predicted number of active projects the company was working on 2 months after the end of November 2012.

Choice D is incorrect. This is the predicted number of active projects the company was working on 4 months after the end of November 2012.

Q6 C

C. $w + 29$

Choice C is correct. It's given that the expression $ww + 29$ represents the area, in square centimeters, of a rectangular painting, where w is the width, in centimeters, of the painting. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the painting is represented by the expression $w + 29$.

Choice A is incorrect. This expression represents the width, in centimeters, of the painting, not its length, in centimeters.

Choice B is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the painting, not its length, in centimeters.

Choice D is incorrect. This expression represents the area, in square centimeters, of the painting, not its length, in centimeters.

Q7 B

B. $gx = \frac{a - 19}{x - 4} + 2$

Choice B is correct. It's given that the graph of $y = gx$ is produced by translating the graph of $y = fx$ 3 units down and 4 units to the right in the xy-plane. Therefore, function g can be defined by an equation in the form $gx = fx - 4 - 3$. Function f is defined by the equation $fx = \frac{a - 19}{x} + 5$, where a is a constant. Substituting $x - 4$ for x in the equation $fx = \frac{a - 19}{x} + 5$ yields $fx - 4 = \frac{a - 19}{x - 4} + 5$. Substituting $\frac{a - 19}{x - 4} + 5$ for $fx - 4$ in the equation $gx = fx - 4 - 3$ yields $gx = \frac{a - 19}{x - 4} + 5 - 3$, or $gx = \frac{a - 19}{x - 4} + 2$. Therefore, the equation that defines function g is $gx = \frac{a - 19}{x - 4} + 2$.

Choice A is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 3 units down and 4 units to the left, not 3 units down and 4 units to the right.

Choice C is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 4 units to the left, not 3 units down and 4 units to the right.

Choice D is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 4 units to the right, not 3 units down and 4 units to the right.

Q8

B

B. 35

Choice B is correct. It's given that the table shows three values of x and their corresponding values of y for the equation $y = 42^x + 3$. It's also given that when $x = 3$ the corresponding value of y is a, and a is a constant. Substituting 3 for x and a for y in the given equation yields $a = 42^3 + 3$, or $a = 35$. Therefore, the value of a is 35.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

D

D. $p = 15{,}0001.04^x$

Choice D is correct. It's given that a model predicts the population of Bergen in 2005 was 15,000. The model also predicts that each year for the next 5 years, the population increased by 4% of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by 1.04. Since the predicted population in 2005 was 15,000, the predicted population 1 year later is $15{,}000(1.04)$. The predicted population 2 years later is this value times 1.04, which is $15{,}0001.041.04$, or $15{,}0001.04^2$. The predicted population 3 years later is this value times 1.04, or $15{,}0001.04^3$. More generally, the predicted population, p, x years after 2005 is represented by the equation $p = 15{,}0001.04^x$.

Choice A is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 0.96 rather than 15,000.

Choice B is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 1.04 rather than 15,000.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by 4% each year.

Q10 B

B. The diver hits the water at 1.6 seconds.

Choice B is correct. It's given that the graph shows the height above the water y, in meters, of a diver x seconds after diving from a platform. The x-intercept of a graph is the point at which the graph intersects the x-axis, or when the value of y is 0. The graph shown intersects the x-axis between $x = 1$ and $x = 2$. In other words, the diver is 0 meters above the water, or hits the water, between 1 and 2 seconds after diving from the platform. Of the given choices, only choice B includes an interpretation where the diver hits the water between 1 and 2 seconds. Therefore, the best interpretation of the x-intercept of the graph is the diver hits the water at 1.6 seconds.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the best interpretation of the maximum value, not the x-intercept, of the graph.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank